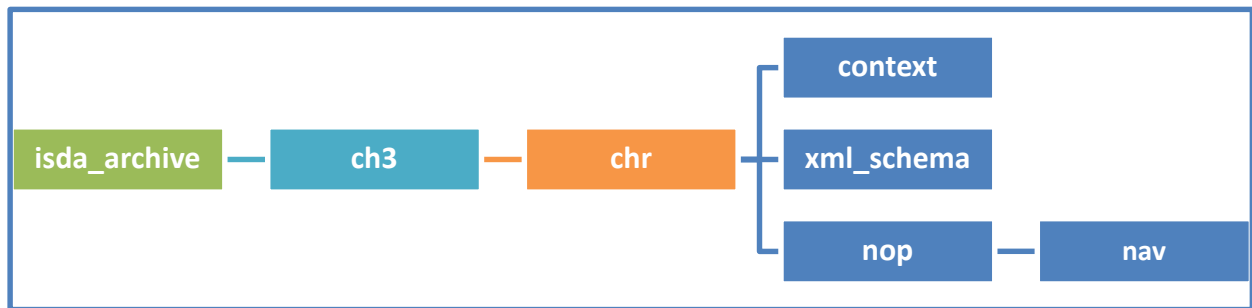


Chandrayaan-3 Rover NavCam PDS4 Data Products Quick User Manual

1. Archive Structure

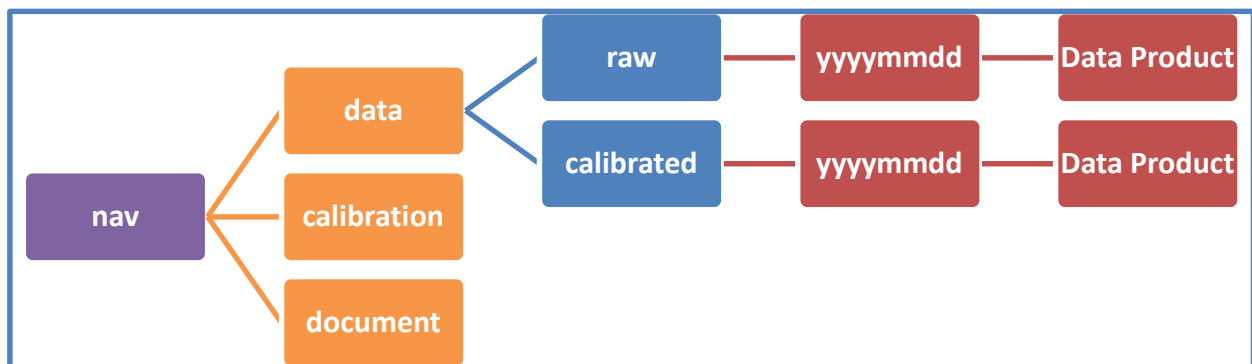
1.1. Mission Archive Structure

At mission level, data is organized as mission id wise bundle. The mission id (ch3-> chandrayaan-3) wise bundle at next level is organized based on the instrument host id (chr-> chandrayaan rover) wise bundle. The instrument host id (nav->navigational camera) bundle at next level is organized based on the mission phase id (nop->normal operations phase). Each mission phase id is having instrument id wise collection. Below figure shows the mission archive structure.



1.2. Instrument Archive Structure

Under each instrument wise data collection, data is organized under each directory defined by PDS4. Data is organized based on the levels of data processing and data products definitions. Data products is organized into year, month and day wise as yyyyymmdd. The year month and day wise directory contains PDS4 data products including PDS4 Label products. Below Figures shows the detail of nav instrument wise collection structure.



2. Data Products Description

The PDS4 Data Products for NavCam are defined based on the defined data processing levels as Raw and Calibrated data products. For each payload observation, PDS4 data products are generated (based on product id & sensor id derived from file name (section 3)) and organized under year month day wise collection directory defined under instrument collection structure shown in section 1.2. Below Figure shows the structure of NavCam Raw and Calibrated Data Product file.

raw/20230828	calibrated/20230828
— ch3_nav_nrl_20230828T0555550000_d_img_d32_010.png	— ch3_nav_ncl_20230828T0555550000_d_img_d32_010.png
— ch3_nav_nrl_20230828T0555550000_d_img_d32_010.xml	— ch3_nav_ncl_20230828T0555550000_d_img_d32_010.xml
— ch3_nav_nrl_20230828T1732040000_d_img_d32_013.png	— ch3_nav_ncl_20230828T1732040000_d_img_d32_013.png
— ch3_nav_nrl_20230828T1732040000_d_img_d32_013.xml	— ch3_nav_ncl_20230828T1732040000_d_img_d32_013.xml
— ch3_nav_nrl_20230828T2026060000_d_img_gnh_014.png	— ch3_nav_ncl_20230828T2026060000_d_img_gnh_014.png
— ch3_nav_nrl_20230828T2026060000_d_img_gnh_014.xml	— ch3_nav_ncl_20230828T2026060000_d_img_gnh_014.xml
— ch3_nav_nrr_20230828T0556380000_d_img_d32_010.png	— ch3_nav_ncr_20230828T0556380000_d_img_d32_010.png
— ch3_nav_nrr_20230828T0556380000_d_img_d32_010.xml	— ch3_nav_ncr_20230828T0556380000_d_img_d32_010.xml
— ch3_nav_nrr_20230828T1732450000_d_img_d32_013.png	— ch3_nav_ncr_20230828T1732450000_d_img_d32_013.png
— ch3_nav_nrr_20230828T1732450000_d_img_d32_013.xml	— ch3_nav_ncr_20230828T1732450000_d_img_d32_013.xml
— ch3_nav_nrr_20230828T2026470000_d_img_gnh_014.png	— ch3_nav_ncr_20230828T2026470000_d_img_gnh_014.png
— ch3_nav_nrr_20230828T2026470000_d_img_gnh_014.xml	— ch3_nav_ncr_20230828T2026470000_d_img_gnh_014.xml

Raw Data Products contains raw data without any correction. Calibrated Data Products contains radiometrically corrected data along with the refined corner coordinates. Below table shows the details of individual data products (with the data formats) for both raw and calibrated.

PDS4 Data Product Levels	Data	Data Formats	Description	Visualization tools available
Raw	Data Raw Image	PNG	The PDS Raw data in png format	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Data Raw Label	XML	The PDS Label Product contains description about the Raw Product.	PDS4 Viewer
Calibrated	Data Calibrated Image	PNG	The PDS Radiometrically corrected data in png format	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Data Calibrated Label	XML	The PDS Label Product contains description about the Calibrated Product.	PDS4 Viewer

3. File Naming Conventions & Formats

This section describes the naming convention used for science data files for the Chandrayaan-3 mission. below shows the detail description about the following file naming conventions designed for the data products.

ch3_<nav>_<mtc>_<YYYYMMDDTHHMMSSsss>_<p>_<prd>_<stn>_mob.fff

Code	Description
ch3	It denotes the Mission name whose value is fixed as ch3
nav	It denotes the Instrument id whose value is fixed as nav
m	It denotes the mission phase name, it can have following value = e (refers to the Earth Bound Phase - ebp) = n (refers to the Normal Operations Phase - nop)
t	It denotes the data type; it can have following values: = r (it denotes raw data) = c (It denotes calibrated data) = d (It denotes derived data)
c	It denotes the imaging mode/camera id/band. It can be one of the following: = l (it denotes left camera of NavCam) = r (it denotes right camera of NavCam)
YYYYMMDDTHHMMSSsss	It denotes the observation/imaging start time in format [YYYYMMDDTHHMMSSsss] YYYY = stand for year MM = stand for month in MM format DD = stand for day in DD format T = stand for time. HH = stand for hour in HH format MM = stand for minute in MM format SSS = stand for seconds in SS format sss = stand for nanosecond in ssss format
p	It denotes the PDS Data Products categories. It value can be any of the following: = d (It denotes data products under data directory) = b (it denotes browse data products under browse directory) = g (it denotes gridded data products under geometry directory)

prd	It denotes the PDS Data Product Name. It value can be any of the following: = img (It denotes the image data product) = brw (It denotes the browse data product) = grd (It denotes the gridded data product) = dtm (it denotes DTM derived product) = oth (it denotes Ortho derived product)
stn	It denotes the station id from which data was acquired. It value can be any of the following: = d32 (ISSDC Bangalore) = d18 (ISSDC Bangalore) = gds (Gold Stone USA) = cnb (Canberra, Australia)
mob	Rover mobility id given by the mission team
fff	It denotes the standard file type extension. It can have any one of the following values: = png (PDS Image Data File) = xml (PDS Detached Label File) = png (PDS Browse Data File) = csv (PDS Geometry Data File) = tif (PDS Geo tiff data)

Below table shows sample file naming, data formats for the instruments mentioned in the above table.

Instrument	File Name	Data Formats
NAV Left	ch3_nav_ncr_20230824T1415420000_d_img_d32_002.png ch3_nav_ncr_20230824T1415420000_d_img_d32_002.xml	PNG XML
NAV Right	ch3_nav_ncl_20230824T1415110000_d_img_d32_002.png ch3_nav_ncl_20230824T1415110000_d_img_d32_002.xml	PNG XML

4. Data Disclaimer

It is to bring to the notice of the user that the Department of Space/Indian Space Research Organization (DOS/ISRO) is not responsible for any dispute/disrepute that may arise due to the use of this dataset. The user should neither redistribute the data to third party nor it be used for any commercial purpose without the knowledge of Department of Space/Indian Space Research Organization (DOS/ISRO).